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Fundamentals of Cryptocurrency Perpetual Futures and Swaps

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Abstract

Perpetual futures (often called perpetual swaps) are the dominant crypto-derivatives instrument. They replicate the economic exposure of a futures contract without an expiry date. They replace maturity-based convergence with a funding mechanism that transfers cash flows between longs and shorts, typically every eight hours. This paper explains how perpetuals evolved from early proposals for non-maturing futures into a standardized crypto market instrument, and why key design choices changed over time. It synthesizes recent theoretical and empirical research on funding design, pricing, and arbitrage intuition, market microstructure, liquidation risk, and regulation. Finally, this study proposes a research agenda organized around funding design, constrained arbitrage, transparency, decentralized exchange design, policy, and legal classification, because recent U.S. and EU developments show that the same economic structure may be characterized as a futures contract, swap, CFD-type instrument, or other derivative depending on statutory definitions, venue design, and supervisory interpretation. This paper proposes the following definition: **a cryptocurrency perpetual is an open-ended, margin-based derivative that gives synthetic long or short exposure to an underlying crypto asset and replaces expiry-based settlement with periodic funding payments that anchor the contract price to a reference spot price.**

Keywords: cryptocurrency derivatives; perpetual futures; perpetual swaps; funding rate; liquidation; market microstructure; expiry-based settlement; periodic funding payments

Introduction

Perpetual futures are the dominant derivative traded in cryptocurrency markets, in terms of both trading volume and participation breadth (He et al., 2024; Gornall et al., 2024). They provide leveraged long or short exposure to an underlying without spot custody and without rolling expiring contracts. This operational simplicity matters because expiring futures fragment liquidity across maturities and impose recurrent rollover costs. Perpetuals remove maturity, but they also remove the terminal convergence that standard futures obtain at expiry. The absence of maturity creates a design problem: the contract price can drift away from the underlying spot price unless another anchoring force exists. Crypto exchanges solve this problem with a funding mechanism that transfers payments between long and short positions as a function of the perpetual–spot spread (Giagkiozis & Said, 2024; He et al., 2024). This paper treats perpetuals as an evolving market design. It traces the historical evolution from early proposals to current centralized and decentralized implementations and links design choices to observed market outcomes. It also identifies open research questions that follow from the instrument’s rapid diffusion into global markets and its emerging policy relevance (Phang, 2023; Dell’Erba & Vianelli, 2025). This paper is part of PGRI’s papers about cryptocurrency perpetuals and behavioral finance (Neubert et al., 2026a; 2026b)

Definitions and Taxonomy

Researchers often use “perpetual futures” and “perpetual swaps” interchangeably, but the terminology reflects market convention rather than a stable legal classification. In crypto, exchanges commonly label the product a “perpetual swap,” while the contract’s economic exposure resembles a futures position with continuous marking-to-market. Giagkiozis and Said (2024) define perpetual swaps as futures contracts with no expiry that use a funding rate to tether

the contract price to a spot index. Ruan and Streltsov (2022) likewise treat perpetual contracts as Shiller-style futures without expiration whose funding mechanism substitutes for maturity convergence. This paper uses “perpetual” as the umbrella term and clarifies subtypes by payout and collateral design.

Legal classification should be separated from economic function, because market labels do not determine whether a perpetual falls within a futures, swap, CFD, or other-derivative category (Chicago Mercantile Exchange Inc., 2026; Commodity Futures Trading Commission, 2026a, 2026b; Directive 2014/65/EU, 2014; European Securities and Markets Authority, 2025). Economically, a perpetual gives synthetic exposure to the underlying asset without requiring spot custody or delivery of the underlying asset (Ackerer et al., 2025; He et al., 2024). The position remains open until offset, liquidation, or termination (Ackerer et al., 2025; He et al., 2024). This structure differs from standard futures because it achieves terminal convergence through expiry-based delivery or cash settlement, whereas perpetuals use funding payments to create continuous convergence incentives (Commodity Futures Trading Commission, 2026a; He et al., 2024). This structure also contains swap-like elements because participants exchange payments based on the value of an underlying asset without receiving ownership of that asset (Commodity Exchange Act, 2018; Commodity Futures Trading Commission, 2023). The instrument is therefore best described economically as a hybrid derivative, while its legal classification remains jurisdiction-specific (Chicago Mercantile Exchange Inc., 2026; Dell’Erba & Vianelli, 2025; European Securities and Markets Authority, 2025).

A consolidated definition must capture the features that drive both pricing and risk. Perpetuals are derivative contracts on an underlying asset that (a) have no contractual maturity, (b) are marked to market through an exchange-defined mark price, (c) require margin collateral

and allow leverage, and (d) include periodic funding payments between longs and shorts that depend on the perpetual–spot spread and often include an interest-rate component. Ackerer et al. (2025) formalize these features in a no-arbitrage setting and show that the funding rule is central to how the contract price remains near the spot price. This definition also distinguishes perpetuals from classic OTC interest rate swaps, even though both involve periodic payments, because perpetuals embed exchange-style variation margin and automatic liquidation.

Taxonomy is necessary because “a perpetual” is not one instrument. The first axis is payoff and collateral currency. Linear perpetuals are typically margined and settled in the quote currency, often a USD stablecoin, and their payoff is linear in the quote currency. Inverse perpetuals are margined and settled in the base asset (e.g., BTC for a BTC/USD contract) and therefore have a nonlinear payoff in fiat terms. Giagkiozis and Said (2024) explain that inverse contracts dominated early because exchanges could operate without holding fiat collateral, but stablecoin adoption shifted demand toward linear contracts. A third subtype is the quanto perpetual, which uses a third currency for margining and settlement, thereby introducing an additional exchange-rate exposure (Ackerer et al., 2025; Atzberger et al., 2024).

A second axis is venue design. Centralized exchanges (CEXs) typically use a limit order book and internal risk engines for margining and liquidation. Decentralized exchanges (DEXs) implement perpetuals via smart contracts and differ in how they form prices and manage risk. Chen et al. (2024) classify DEX perpetuals into models such as oracle pricing and virtual automated market makers (VAMMs), which shift price formation from an order book to algorithmic inventory rules. This choice of venue matters because it affects who provides liquidity, how liquidations are executed, and how transparent collateralization is.

Historical Evolution: From Concept to Crypto Market Standard

Perpetual futures originated as a proposed solution to the maturity and illiquidity frictions of standard futures. Shiller (1993) proposed perpetual futures for settings in which cash settlement requires a robust index and maturity cycles complicate hedging and price discovery. The core idea was to replace delivery at maturity with repeated cash settlements tied to a reference index, allowing traders to maintain exposure without rollovers. Crypto markets implemented this concept for a different reason. They needed a continuously tradable, high-leverage instrument that can operate on venues with heterogeneous collateral constraints and limited access to banking infrastructure (Giagkiozis & Said, 2024; Dell’Erba & Vianelli, 2025).

BitMEX introduced the first crypto perpetual swap in 2016 and established a template that other venues later adopted. Giagkiozis and Said (2024) document that the first perpetual swap was an inverse contract, with margin and profit and loss paid in the base asset. This design aligned with early market constraints, as many exchanges did not accept fiat deposits and could therefore only collateralize with crypto. Ruan and Streltsov (2022) and Chen et al. (2024) also date the introduction of crypto to 2016 and emphasize that the absence of expiry made perpetuals operationally attractive for traders seeking continuous exposure.

Funding design evolved in identifiable stages. Early tethering used the USD–Bitcoin lending-rate differential from the Bitfinex lending market, and positions exchanged these cash flows every eight hours (Giagkiozis & Said, 2024). As cross-exchange arbitrage strengthened, exchanges moved to index-based funding formulas that compare the perpetual price to a multi-venue spot index and set funding to incentivize convergence trades when futures deviate

from the spot price. This design reduced the risk of single-venue manipulation but shifted attention to index construction and mark-price governance (He et al., 2024).

Exchanges then added guardrails that reflect the system's interaction with leverage and liquidations. He et al. (2024) document the widespread use of funding “clamps,” meaning funding behaves differently when the futures–spot gap is small versus large. This design attempts to reduce destabilizing feedback loops when prices move sharply and to prevent mechanically extreme funding rates. The same period saw a broad migration from inverse to linear contracts. Giagkiozis and Said (2024) link this shift to stablecoin adoption and to traders' preference for accounting in fiat units. Alexander et al. (2023) provide related evidence that product choice correlates with liquidation risk, as coin-margined and stablecoin-margined contracts exhibit different collateral dynamics.

The latest evolutionary step is the expansion of perpetuals into decentralized finance. Chen et al. (2024) argue that post-2022 demand for transparency increased the adoption of DEX perpetuals. DEX implementations replace centralized custody with smart-contract custody and often replace order books with oracle pricing or VAMM mechanisms. This shift changes the economic roles of liquidators and liquidity providers and introduces oracle and smart-contract risk as part of the derivatives lifecycle.

Contract Mechanics: Lifecycle, Funding, Margin, and Liquidation

Perpetual mechanics link four processes: position entry and marking, funding payments, margining, and liquidation. He et al. (2024) describe the lifecycle as a continuous mark-to-market of positions plus discrete funding transfers that depend on the futures–spot spread. Exchanges compute mark prices (often index-based) to reduce manipulation of last-traded prices and to standardize liquidation triggers. This approach makes the mark price a

governance variable because it determines when positions get liquidated, not only how profits and losses are displayed.

Funding payments operationalize the convergence pressure that expiry would otherwise provide. When the perpetual trades above spot, funding is typically positive, and longs pay shorts. When the perpetual trades below spot, funding is typically negative, and shorts pay longs. This payment rule creates an incentive for arbitrageurs to take the opposite side of the mispricing, but it does not guarantee convergence at a fixed date. He et al. (2024) therefore frame arbitrage in perpetuals as “random-maturity” because the unwinding time is endogenous and uncertain. Ackerer et al. (2025) show that different funding specifications imply different no-arbitrage prices and can even generate designs that keep the basis near zero in frictionless settings. The operational implication is that “the funding rate” is part of contract design, not merely a market outcome.

Margin and liquidation transform this design into a leveraged retail product. Many major venues offer leverage of around $100\times$ – $125\times$, and some platforms advertise even higher values (Giakiozis & Said, 2024). High leverage implies that small adverse moves trigger forced liquidations, which exchanges execute automatically. Cheng et al. (2021) quantify this mechanism for BitMEX perpetuals and show that liquidated traders use, on average, very high leverage. Their results imply that margin schedules can materially affect the incidence of liquidation and market stability. Alexander et al. (2023) extend this risk perspective by modeling optimal hedging under conditions in which liquidation is possible, and leverage is self-selected. They show that optimal hedge ratios depend not only on covariance structure but also on liquidation loss aversion and margin constraints.

Transparency metrics, such as open interest, should support margin risk assessment, but crypto venues do not standardize their reporting. Giagkiozis and Said (2024) provide evidence that some major exchanges misreport open interest and show how changes in open interest should mechanically relate to traded volume. This matters because open interest is a lower bound on collateral needs, so misreporting weakens external solvency checks. The mechanics imply that market integrity depends on both contract design and data quality.

Pricing Intuition and Arbitrage: Why the Basis Persists

Perpetual pricing differs from standard futures because maturity-based convergence is replaced by continuous convergence pressure. Akerer et al. (2025) derive no-arbitrage prices for linear, inverse, and quanto perpetuals and show that the funding rule pins down the relationship between spot and perpetual prices. He et al. (2024) provide a complementary framing that emphasizes random-maturity arbitrage. They show that in frictionless markets, the perpetual price is proportional to the spot price, with the proportionality constant depending on interest rates and the funding function. This result explains why basis risk persists even when arbitrage is active: the contract does not enforce $F_t = S_t$, and the funding mechanism itself can create a nonzero wedge.

Empirical work documents persistent but time-varying deviations around these benchmarks. He et al. (2024) report that deviations in crypto markets are larger than in traditional currency markets and that cryptocurrencies co-move, consistent with shared constraints on arbitrage capital. They also find that deviations diminish over time, which is consistent with market maturation and increased competition among arbitrageurs. Gornall et al. (2024) provide complementary evidence that perpetuals reduced extreme dislocations relative to quarterly futures by replacing one-time maturity convergence with continuous funding-based convergence.

Their analysis implies that contract structure can change the risk profile of basis trades, which matters for market stability when leveraged arbitrageurs provide liquidity.

The canonical arbitrage trade is conceptually simple: buy spot and short the perpetual when the perpetual is rich, and finance the position with leverage while collecting funding. In practice, this trade is risky because liquidation risk and funding volatility can dominate short horizons (Cheng et al., 2021). Alexander et al. (2024) show that cross-market arbitrage in crypto derivatives can remain profitable even under conservative transaction-cost assumptions, implying incomplete arbitrage. In perpetuals, the same logic implies that “arbitrage bounds” rather than a single no-arbitrage price can describe feasible trading, especially for traders facing higher fees, margin constraints, and operational risk (He et al., 2024). Persistent basis is therefore an equilibrium outcome of funding design, transaction costs, and liquidation constraints, not a standalone anomaly.

Market Quality and Microstructure: Funding Cycles and Information

Perpetuals affect market quality by coupling derivatives and spot markets through funding payments and convergence trading. Ruan and Streltsov (2022) document a U-shaped intraday pattern over the eight-hour funding cycle. They find that spot trading volume increases around funding times, but quoted spreads widen at the same time. They interpret the pattern as increased informed trading around funding settlement, which raises adverse selection risk for market makers. This evidence clarifies why higher volume does not necessarily imply better liquidity when the additional volume is informationally “toxic.”

Perpetuals also affect liquidity across contract types and under market stress. Gornall et al. (2024) show that perpetual futures are more liquid than quarterly futures and remain more liquid during periods of market instability. They also document that quarterly futures can

overshoot spot price moves during large events, increasing drawdowns for basis arbitrage. This implies that perpetual design can improve crisis-time liquidity by reducing basis volatility. Ruan and Streltsov (2022) add nuance by showing that even if perpetuals improve derivatives-side liquidity, they can increase adverse selection risk in spot markets around settlement events.

Data quality is itself a microstructure issue because it affects inference and monitoring. Giagkiozis and Said (2024) show that open interest misreporting can be detected by comparing high-frequency changes in open interest with traded volume. The result implies that researchers and regulators should treat reported open interest as a potentially noisy state variable unless exchanges disclose consistent definitions and timestamps. It also matters to market participants because open interest is widely used as a proxy for sentiment and risk.

Regulation and Policy: Leverage, Consumer Risk, and Cross-Border

Fragmentation

Regulation of crypto-derivatives varies across jurisdictions and often targets retail leverage and consumer losses. Phang (2023) compares approaches in the United Kingdom, the European Union, and Singapore, documenting divergent policy choices ranging from prohibitions on retail crypto-derivatives trading to regulated access through approved venues. This variation matters for perpetuals because the product's defining features, high leverage, continuous trading, and frequent liquidation, amplify consumer protection concerns. The policy tradeoff is direct: bans reduce retail harm but may push trading offshore, while permissive regimes require robust supervision of platforms, custody, margining, and market integrity.

Regulatory fragmentation also creates challenges for enforcement and systemic risk. Dell'Erba and Vianelli (2025) analyze crypto-derivatives regulation across major jurisdictions and argue that inconsistencies enable regulatory arbitrage. They emphasize that the settlement

method, eligible underlyings, and policy stance create divergent perimeter outcomes even when contracts replicate the structure of traditional derivatives. For perpetuals, this implies that similar instruments may be treated as futures, swaps, or CFDs, depending on local classification and marketing practices, complicating cross-border supervision.

Legal Classification: Futures, Swaps, CFDs, and Other Derivatives

In the United States, the legal perimeter of perpetuals became contested in 2026. The CFTC approved KalshiEX's BTCPERP, a bitcoin perpetual contract, for listing as a futures contract on a designated contract market (Commodity Futures Trading Commission, 2026a). Its policy statement defines perpetuals as derivative contracts without fixed expiration dates that use periodic funding rather than expiry to maintain price parity with spot (Commodity Futures Trading Commission, 2026b). The Kalshi order distinguishes this design from standard futures, which converge toward spot at expiry through delivery or cash settlement (Commodity Futures Trading Commission, 2026a). CME disputes this classification and argues that crypto perpetuals fit the CEA swap definition because they exchange payments based on a commodity's value, transfer price risk, do not transfer ownership, and lack expiry or delivery (Chicago Mercantile Exchange Inc., 2026; Commodity Exchange Act, 2018). This dispute does not settle the legal nature of perpetuals. It shows that classification depends on statutory interpretation, agency policy, product design, and judicial review (Chicago Mercantile Exchange Inc., 2026; Commodity Futures Trading Commission, 2026a, 2026b).

In the European Union, the safer approach is functional rather than label-based. MiFID II Annex I Section C includes futures, swaps, financial contracts for differences, and other derivative contracts within the category of financial instruments (Directive 2014/65/EU, 2014). ESMA's crypto-asset guidelines require a case-by-case, substance-over-form, and

technology-neutral assessment (European Securities and Markets Authority, 2025). A crypto perpetual may therefore qualify as a MiFID II financial instrument, especially as a CFD-type or other derivative contract, without being a classical future or classical swap in the narrow sense (Directive 2014/65/EU, 2014; European Securities and Markets Authority, 2025). This approach supports a functional analysis of perpetuals based on economic exposure, funding design, venue structure, and regulatory perimeter, rather than on market label alone (Dell’Erba & Vianelli, 2025; European Securities and Markets Authority, 2025).

The empirical risk channel that regulators confront is leverage-induced liquidation cascades. Cheng et al. (2021) document substantial forced liquidations in BitMEX perpetual markets and recommend higher margin requirements to reduce the probability of margin calls. Alexander et al. (2023) show that liquidation risk alters optimal hedging behavior, implying that the product can fail as a risk-management tool if leverage is set too aggressively. These findings support a policy emphasis on margin standards, disclosure, and data integrity rather than on product labels alone.

Research Agenda

A fundamentals paper should end with a research agenda that converts design facts into testable questions. Existing work identifies open issues in pricing and regulation, but it remains fragmented across finance, computer science, and legal scholarship. Building on the evidence summarized above, we propose eight research themes, each with testable questions.

1. Funding design and endogeneity. How do exchanges set clamps, caps, and index rules in response to order flow and liquidation risk, and do these design choices reduce basis volatility without increasing manipulation (Ackerer et al., 2025; He et al., 2024)?

2. Constrained arbitrage and basis dynamics. Which frictions (fees, margin rules, capital constraints, or market segmentation) explain cross-coin comovement of basis and its time trend, and do these frictions differ between CEX and DEX venues (Gornall et al., 2024; He et al., 2024)?

3. Microstructure around funding events. Does informed trading concentrate at funding times because funding aggregates information, because liquidations cluster mechanically, or because arbitrageurs rebalance inventory? Researchers can test this by decomposing volume into arbitrage versus directional trades around settlement windows (Ruan & Streltsov, 2022).

4. Liquidation cascades and optimal margin. How should exchanges set dynamic initial and maintenance margin schedules that internalize fat tails and asymmetric payoffs, and how do these schedules change welfare across hedgers, speculators, and market makers (Cheng et al., 2021; Alexander et al., 2023)?

5. Transparency, open interest, and solvency monitoring. Which reporting standards would make open interest and liquidation data sufficiently reliable for external monitoring, and can cryptographic proof-of-reserve and proof-of-liability mechanisms be integrated into derivatives open-interest reporting (Giagkiozis & Said, 2024)?

6. DeFi perpetual design and oracle risk. How do oracle pricing and VAMM mechanisms change adverse selection, liquidation efficiency, and price impact relative to limit-order-book CEXs, and which design features reduce oracle manipulation risk (Chen et al., 2024)?

7. Regulation and market migration. How do regulatory restrictions on retail access shift trading toward offshore venues and DEXs, and do these migrations increase or decrease consumer losses and systemic risk? Comparative event studies across jurisdictions can identify substitution effects (Phang, 2023; Dell’Erba & Vianelli, 2025).

8. Legal classification and regulatory design. How do legal classifications affect contract design choices such as stated expiry, funding formulas, margin rules, venue registration, retail access, and disclosure? Future research should test whether regulatory labels map onto economic risk or instead encourage economically equivalent products with different legal forms (Chicago Mercantile Exchange Inc., 2026; Coinbase Derivatives, LLC, 2026; Commodity Futures Trading Commission, 2026a, 2026b; Commodity Futures Trading Commission, Division of Market Oversight, 2026).

Conclusion

Perpetual futures evolved through a sequence of design choices that solved a specific coordination problem: how to offer leveraged futures exposure without maturity-based convergence. Funding mechanisms, mark-price rules, and liquidation engines jointly replace the expiry anchor. The historical trajectory from Shiller's proposal to BitMEX's 2016 implementation and to today's CEX and DEX variants shows that market structure and regulation shaped contract form. Recent theory formalizes how funding rules determine no-arbitrage relationships, while empirical work shows that funding cycles reshape spot liquidity and that leverage and liquidation remain central risks. The research agenda points to the next step: treat perpetuals as governed market infrastructure and measure how design changes map into welfare, stability, and investor outcomes. The emerging U.S. dispute over whether crypto perpetuals are futures or swaps reinforces this conclusion, as it shows that perpetuals cannot be understood solely by market labels and should instead be evaluated as a governed market infrastructure.

Defined functionally, a **cryptocurrency perpetual is an open-ended, margin-based derivative that gives synthetic long or short exposure to an underlying crypto asset and**

replaces expiry-based settlement with periodic funding payments that anchor the contract price to a reference spot price (Ackerer et al., 2025; Commodity Futures Trading Commission, 2026a, 2026b; He et al., 2024). This definition remains legally neutral: it explains the product's economic design while leaving formal classification as a future, swap, CFD-type instrument, or other derivative to statutory interpretation, contract specifications, venue architecture, and supervisory judgment (Chicago Mercantile Exchange Inc., 2026; Directive 2014/65/EU, 2014; European Securities and Markets Authority, 2025).

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Declarations

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Data Availability and Supplementary Material

All data generated and analyzed during this study and supplementary material are available upon reasonable request.

Prior Publication

The author confirms that this research has not been published previously and is not under consideration for publication elsewhere.

Ethics Statement

This study complies with the ethical guidelines of the European Code of Conduct for Research Integrity and adheres to the GDPR requirements for data protection.

Responsible AI Ethics Statement

This study used artificial intelligence (AI) tools to support tasks such as identifying relevant literature, analyzing datasets, translating, and editing textual content. These tools were employed solely to enhance efficiency, and their outputs were critically reviewed to ensure alignment with research objectives. The use of AI adheres to ethical principles outlined in the Regulation (EU) 2024/1689, the OECD Recommendation of the Council on Artificial Intelligence, and UNESCO's Recommendation on the Ethics of Artificial Intelligence, emphasizing transparency, fairness, and accountability. The authors made all final decisions and retain full responsibility for the integrity, rigor, and conclusions of this research.

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